Proceedings of the ASME 2021
Verification and Validation Symposium
VVS2021
May 19-20, 2021, Virtual, Online

VVS2021-65326

ERROR QUANTIFICATION OF NUSSELT NUMBER ANALYSIS IN MINIATURE HEAT SINKS: VERIFICATION AND VALIDATION ASSESSMENT

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ABSTRACT

Solution verification is performed to quantify the numerical uncertainty of Nusselt numbers in micro-scale heat sinks obtained from 3-D numerical simulations. A numerical procedure is first developed to calculate local and average Nusselt numbers along 4 different miniature heat sinks. Validation process is performed by comparing the obtained numerical results experimental data. Fairly good agreement between numerical results and experimental data confirms the reliability and accuracy of the proposed numerical procedure. The observed order of accuracy for water flow in a micro-tube with constant uniform heat flux is 1.81 while the observed order of accuracy for conjugate heat transfer of water flow within a microchannel heat sink is estimated as 1.2. The numerical uncertainty for local Nusselt numbers within the investigated microchannel heat sink is estimated to be 0.13.

1. INTRODUCTION

Miniature and micro-scale heat sinks can provide high heat dissipation rates in the range of 50-100 $\mathbf{W/cm^2}$ due to their large surface area to volume ratios. Experimental Nusselt number estimation of different coolants within miniature heat sinks has attracted researchers' attention since 1990s. The micro-scale channel geometries in the past studies cover a wide range of shapes from triangular to tube and rectangular ones [1–5]. The early scientific research was dedicated to assessing the reliability and applicability of the Navier-Stokes equation to predict the fluid flow in small-scale devices such as microtubes/channels heat sinks [6–8]. The idea was to verify if the friction factor and Nusselt number within such small devices follow the conventional theory applied to macro-scale tubes and channels. Reported experimental data on Nusselt number in micro-scale heat sinks are almost in agreement with theoretical and experimental values obtained for the laminar flows in macro-scale heat exchangers [9,10]. El-Genk and Yang [11] studied analytically and numerically the effect of slip boundary condition on pressure drop of isothermal water flow within microtubes with diameters of 100-1000 µm. Results showed that the effect of slip boundary condition on friction factor is noticeable in microtube with diameter less than 200 µm. In another study, El-Genk and Pourghasemi [12] performed analytical analyses combined with 3D numerical simulation to verify the effect of liquid slippage on friction factor within microchannels. It was observed that the small slip length of 2 µm reduced the friction factor by almost 30% in microchannel with hydraulic diameters less than 1000 µm. Micro-scale heat sinks have large surface area to volume ratios and provide high convective heat transfer coefficients desirable in heat management of compact microelectronic packages. Micro-scale heat sinks can handle heat dissipation rates as high as 50-100

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$\mathbf{W/cm^2}$ while keeping the working temperature of the microelectronic components in the recommended safe range by manufacturers.

Despite the large number of conducted research work, there are still significant uncertainties in the available experimental data of Nusselt number in miniature heat sinks. These reported large uncertainties in the experimental data can play a major role in validation process for numerical studies in miniature heat sinks Wu and Cheng [13] conducted experiments to investigate convective heat transfer of water within the microchannels with triangular and trapezoidal cross sections. They also observed that below a critical Reynolds number (Re = 100), the Nusselt number rapidly increases and then its growth rate reduces beyond that threshold. Kim [14] experimentally investigates the heat transfer within 10 different rectangular microchannels using water as working fluid. The experimental results reveal that regardless of channels aspect ratio, the average Nusselt number within the channel follows two different regimes. It was observed that there was a critical Reynolds number (Re_{cr} = 180) below that the average experimental Nusselt numbers were lower than the theoretical value and increased rapidly as the Reynolds number got larger than the Re_{cr}. Moreover, beyond the critical Reynolds number, the Nusselt number exceeded the theoretical value but grew slowly with Reynolds number. The reported uncertainty range in the experimental data on Nusselt number within miniature heat sink can be as high as 20% while some studies have shown up to 15% difference between obtained results from numerical simulation and reported experimental data [9,12]. The objective of this research work is to conduct 3-D numerical simulations to quantify the error in the calculation of Nusselt number within miniature heat sinks. Verification procedures are applied to assess the accuracy and reliability of the implemented numerical procedure to predict the Nusselt number within miniature heat sinks. The effect of conjugate heat transfer on Nusselt number calculation during the experiments in miniature heat sinks will be assessed using the implemented numerical procedure.

2. NUMERICAL PROCEDURE

In the numerical model used in this work, continuity, momentum, and the energy equations are solved with mass flow inlet and pressure outlet boundary conditions. The governing equations are discretized using a finite volume method on a collocated grid scheme. Discretized equations are solved using SIMPLE solver schemes while a second-order upwind interpolation scheme is applied to convection terms. At the interface between solid base material and microchannel walls the continuity of heat flux and temperature are imposed using the following boundary conductions, equation (1),[15].

$$T_f = T_s; \qquad k_f \nabla T_f = k_s \nabla T_s \tag{1}$$

The following procedure is used to calculate the local and average Nusselt numbers along investigated miniature heat sinks in this work. At any location z along the heat sink, the average fluid bulk temperature, average wall temperature and average wall heat flux are calculated through equations (2)-(4). x,y,z are the spatial variables while z is along the miniature heat sink.

$$T_b(z) = \frac{\int \rho u c_p T_f dA}{\int \rho u c_p dA} \tag{2}$$

$$T_w(z) = \frac{\{\oint T_w(x,y,z) ds\}_{walls}}{\{\oint ds\}_{walls}} \tag{3}$$

$$q_w(z) = \frac{\{\oint q_w(x,y,z) ds\}_{walls}}{\{\oint ds\}_{walls}} \tag{4}$$

The average local Nusselt number at any location z, along the microchannel/microtube (heat sink) is then calculated through equation (5).

$$Nu(z) = \frac{q_w(z)\, D_h}{k_b(z)\,[T_w(z) - T_b(z)]} \tag{5}$$

Where, D_h is the heat sink microchannel/microtube hydraulic diameter and $k_b(z)$ is the fluid thermal conductivity evaluated at the local fluid bulk temperature. The average Nusselt number for whole heat sink is calculated by taking the integral of local Nusselt number over the heat sink length of L, using equation (6).

$$Nu_{ave} = \frac{\int_0^L Nu(z) dz}{L} \tag{6}$$

Figure 1 and 2 illustrate the calculated local average Nusselt numbers along a microtube at uniform heat flux and constant wall temperature boundary conditions, respectively. Local average Nusselt numbers are calculated using equations (1)-(5).

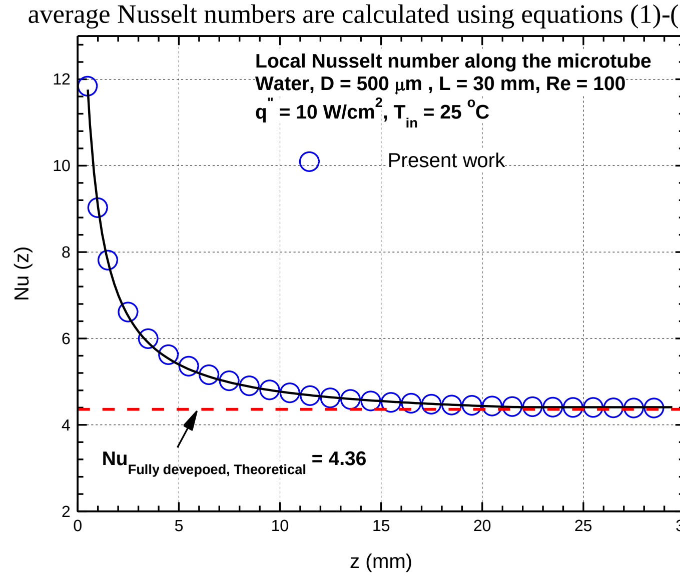

FIGURE 1: LOCAL NUSSELT NUMBERS ALNOG A MICROTUBE AT UNIFORM HEAT FLUX

In both cases, water is used as the coolant within a microtube of diameter 500 µm and length of 30 mm at applied uniform heat flux of 10 W/cm^2 and constant uniform wall temperature of 80 ºC. The local Nusselt number is high in the region near the microtube inlet as the liquid flow is still hydrodynamically and thermally developing. The calculated local Nusselt numbers decrease and approach to the fully developed theoretical values of 4.36 and 3.66 as water moves toward the microtube outlet.

Therefore, the presented numerical results in Figs. 1 and 2 confirm the accuracy and reliability of the implemented numerical approach (equations (2)-(6)) to study flow and heat transfer in miniature heat sinks.

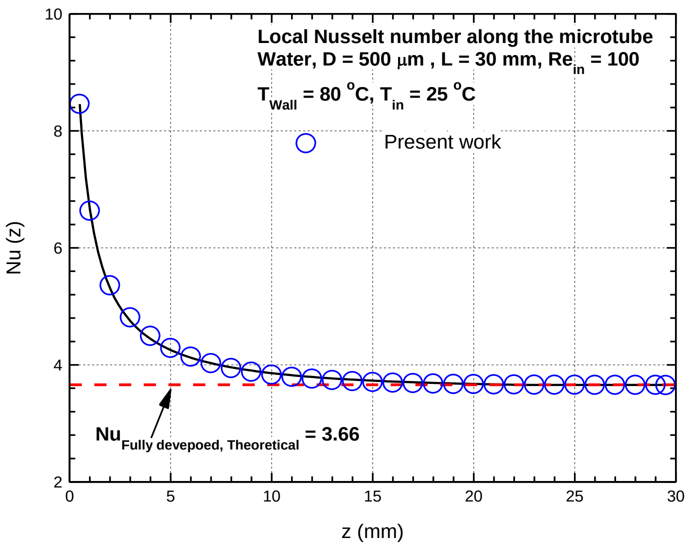

FIGURE 2: LOCAL NUSSELT NUMBERS ALONG A MICROTUBE AT CONSTANT WALL TEMPEARTURE

Table 1 shows the details of the 3 mesh configurations used to obtained order of accuracy of the proposed numerical procedure for local Nusselt number calculation at constant uniform heat flux and wall temperature boundary conditions. L_2 norm is used to calculate the discretization error between results of local Nusselt numbers in two different mesh configurations selected from Table 1.

TABLE 1: MESH GRID CONFIGURATIONS UTILIZED TO OBTAINED ORDER OF ACCURACY FOR WATER FLOW WITHIN THE INVESTIGATED MICROTUBE

Mesh Grid	Mesh elements (Millions)	Inflation layer/ first layer thickness	Average size of mesh elements (µm)
Coarse	0.05975	3 / 4.4 µm	46.2
Intermediate	0.471	6 / 2.2 µm	23.2
Fine	3.31	10 / 1.1 µm	12.1

Figure 3 shows the obtained local Nusselt numbers for coarse and fine mesh grid configurations at constant uniform heat flux boundary condition. L_2 norms of obtained local Nusselt number values along the investigated microtube for presented three mesh configurations in Table 1 are used to calculate the observed order of accuracy. Error is defined as the difference between true value and obtained numerical results. True value is always considered the results obtained from the finer mesh grid configuration. For example, between Coarse and intermediate mesh configuration in Table 1, obtained results from intermediate mesh grid configuration is considered as true value in error calculation. Average size of mesh elements in each mesh configuration is obtained using the microtube volume and total number of mesh elements. The calculated observed order of accuracy using the average size of mesh elements shown in Table 1 and the local Nusselt numbers shown in Figure 3 is 1.81. As it can be inferred from Figure 3, the difference between local Nusselt numbers in coarse and fine mesh grid configuration is small. This is mainly due to the implemented inflation layers at microtube wall. The first layer thickness of implemented inflation layers in coarse and fine mesh grid configurations are 4.4 and 1.1 micron (Table 1). This small inflation layers capture accurately the temperature and velocity gradient near the microtube wall even in coarse mesh grid configuration. Therefore, the difference between the local Nusselt numbers in coarse and fine mesh configuration shown in Figure 3 is small while the observed order of accuracy is 1.81.

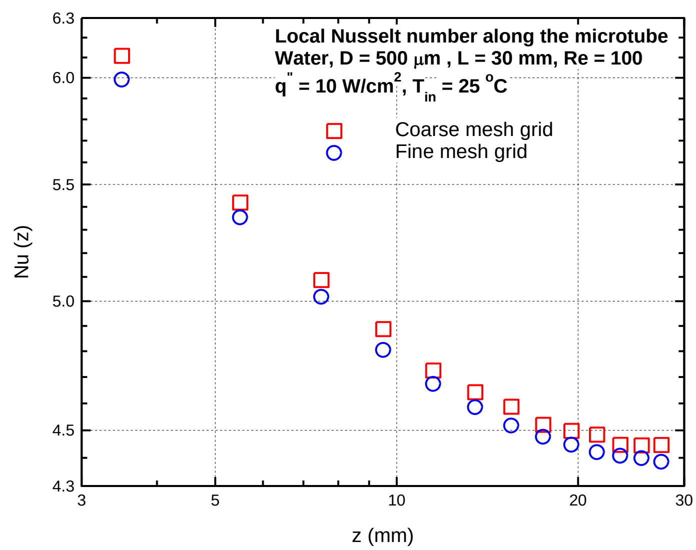

FIGURE 3: LOCAL NUSSELT NUMBERS ALONG A MICROTUBE FOR CORASE AND FINE MESH GRID CONFIGURATION AT UNFORM HEAT FLLUX

Micro-scale heat sinks are directly attached to the hot surface to remove heat from it. Therefore, heat first diffuses through the base solid section of the micro-scale heat sinks and then it is transferred to the coolant passing through small-scale channels resulting in a conjugate heat transfer problem.

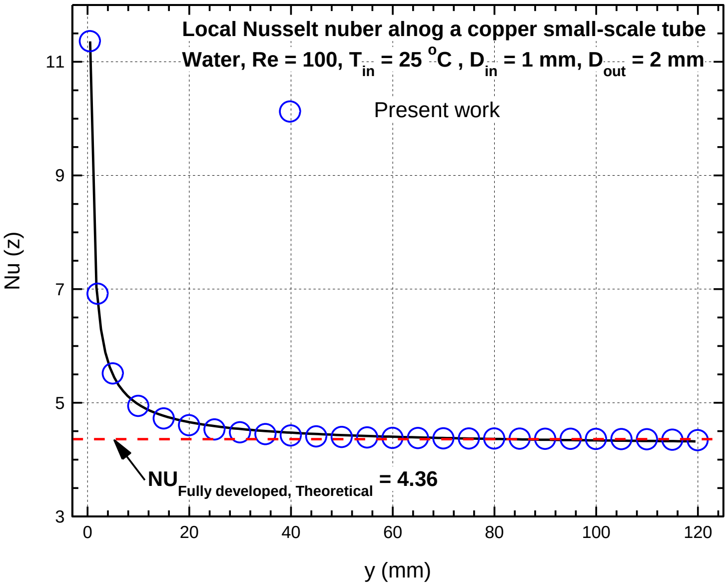

FIGURE 4: LOCAL NUSSELT NUMBERS ALONG A MILLI-TUBE WITH CONJUGATE HEAT TRANSFER AND UNIFROM HEAT FLUX BOUNDARY CONDITION

further numerical simulation is performed to ensure that the implemented numerical procedure using equations (2) to (6) is reliable and accurate in case of conjugate heat transfer problem within miniature heat sinks. A small-scale copper tube with inner diameter of 1 mm and outer diameter of 2 mm is modeled. The thickness of the small-scale copper tube is 2 mm. Constant uniform heat flux of 10 W/cm^2 is applied to the outer wall of the investigated tube. Figure 4 illustrates the obtained numerical results for local Nusselt numbers for water flow at Reynolds number of 100. Good agreement between obtained Nusselt numbers and fully developed theoretical value of 4.36 further confirms the reliability and accuracy of proposed numerical approach to study conjugate heat transfer in miniature heat sinks.

3. RESULTS AND DISCUSSION

Numerical simulations are conducted in this section to investigate heat transfer of water in a copper microchannel heat sink as shown in schematically in Figure 5 to assess the numerical uncertainty of the implemented numerical procedure (equations 2-6). The microchannel has width of W = 206 μm, height of H = 408 μm and length of L = 25 mm in compliance with reported experimental set-up [9]. The aspect ratio of the microchannel which is defined as the ratio of its height to its width ($\alpha = H/W$) is 1.98. The base silicon section has thickness of HS = 2 mm. The applied uniform heat flux at the bottom surface of silicon heat sink is 50 W/cm^2 and the water inlet temperature is 20 ºC. All thermophysical properties of the working fluid, water, have been considered to vary with temperature. Half of the geometry shown in the Figure 5 is modelled in numerical simulations due to symmetry of the microchannel heat sink over the vertical axis passing through middle of microchannel bottom wall (see Fig. 5). Uniform mass flow rate has been specified for channel inlet and pressure outlet boundary condition is adopted for channel outlet. The Reynolds number varies from 300 to 850.

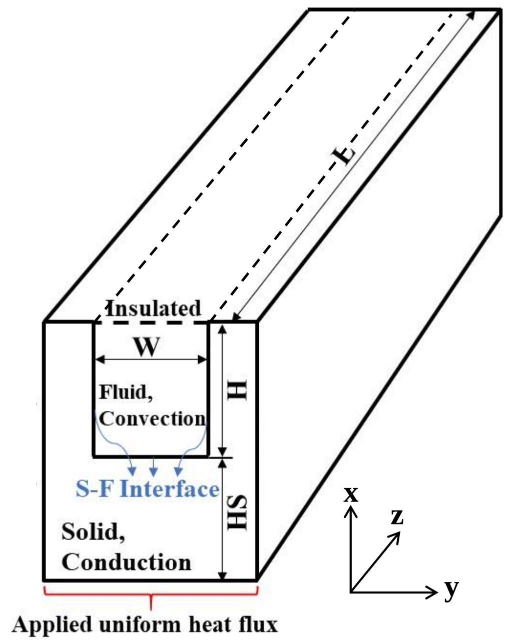

FIGURE 5: SCHEMATIC OF INVESTIGATED COPPER MICROCHANNEL IN NUMERICAL SIMULATIONS

Table 2 shows the sensitivity of the obtained numerical results of Nusselt number to mesh grid refinement in investigated microchannel heat sink at Reynolds of 800. Several inflation layers are applied to the region near microchannel walls to accurately capture the velocity and temperature gradients. As it can be seen from Table 2, obtained Nusselt numbers vary slightly by 0.47% between mesh grids of 1 and 2 configurations, respectively. Moreover, this verifies that the implemented inflation layers are fine enough to precisely model temperature gradient near microchannel walls. Therefore, all numerical simulations in this work are performed using fine mesh grid configuration described in Table 2 to make a compromise between accuracy of the obtained results and computational cost.

TABLE 2: SENSITIVITY OF OBTAINED NUSSELT NUMBERS OF WATER FLOW IN A COPPER MICROCHANNEL HEAT SINK TO NUMERICAL MESH GRID REFINEMENTAT INLET REYNOLDS NUMBER OF 750

Mesh Grid	Mesh elements (Millions)	Inflation layer/ first layer thickness	Nu_{ave}	Diff
1	0.72	5 / 2 μm	6.68	---
2	1.92	8 / 1.33 μm	6.83	~ 2.2 %
3	4.33	10 / 0.9 μm	6.92	~ 1.32 %
4	7.8	12 / 0.6 μm	6.965	~ 0.67 %
5	12.81	14 / 0.4 μm	7.0	~ 0.47 %

Figure 6 compares the obtained average Nusselt numbers from numerical simulations with experimental data. As it can be seen from the Figure 6, in agreement with experimental data, the obtained Nusselt numbers from numerical simulations increase by increasing the inlet Reynolds number. The numerical results follow the trend of experimental data while they are consistently lower. Obtained numerical results shown in Figure 2 confirm the reliability of implemented numerical procedure as well as utilized mesh grid configuration to study fluid flow and heat transfer within micro-scale heat sinks.

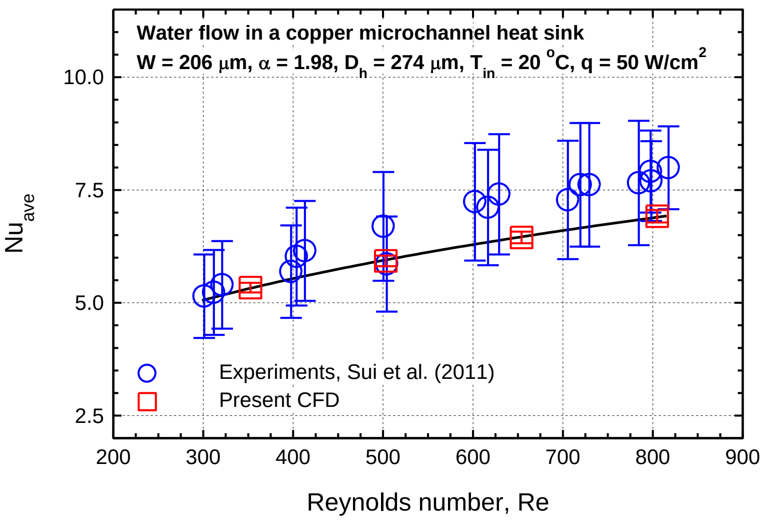

FIGURE 6: COMPARISON BETWEEN OBTAINED NUMERICAL RESULTS AND EXPERIMENTAL DATA OF AVERAGE NUSSELT NUMBERS

Figure 7 shows the empirical distribution function for experimental data and obtained numerical data for average Nusselt numbers shown in Figure 6.

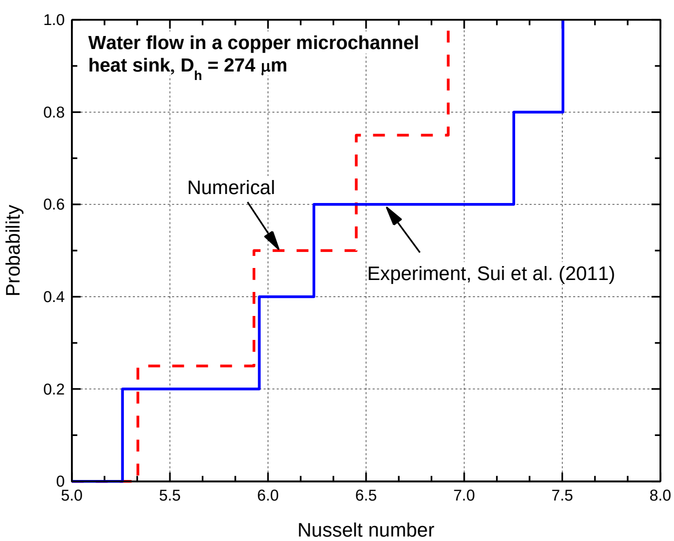

FIGURE 6: EMPIRICAL DISTRIBUTION FUNCTIONS FOR EXPERIMENTAL DATA AND NUMERICAL RESULTS SHOWN IN FIGURE 6

The horizontal axis for experiment is the average value of the Nusselt numbers reported at Reynolds numbers of 300, 400, 500, 600, 700 and 800 in Figure 6. For numerical EDF, obtained average Nusselt numbers at Reynolds number of 350, 500, 650 and 800 are used. The area between the two calculated EDFs in Figure 7 resembles the difference between numerical results and experimental data. Area validation metric, d is equal to 0.3193 for the current presented data of average Nusselt numbers in Figure 6.

Local Nusselt numbers for different mesh grid configurations in Table 2 are used to calculate the observed order of accuracy for water flow within the investigated microchannel. Figure 8 shows the obtained observed order of accuracy. As it can inferred, the observed order of accuracy is around 1.2 while the formal order of accuracy is 2. Conjugate heat transfer occurs within microchannel heat sink while the mesh is non-uniform both in solid and fluid sections in the computational domain with finest mesh elements at the interface of microchannel walls. Conjugate heat transfer and non-uniform mesh configuration cause the observed order of accuracy to be lower than the formal one which is 2.

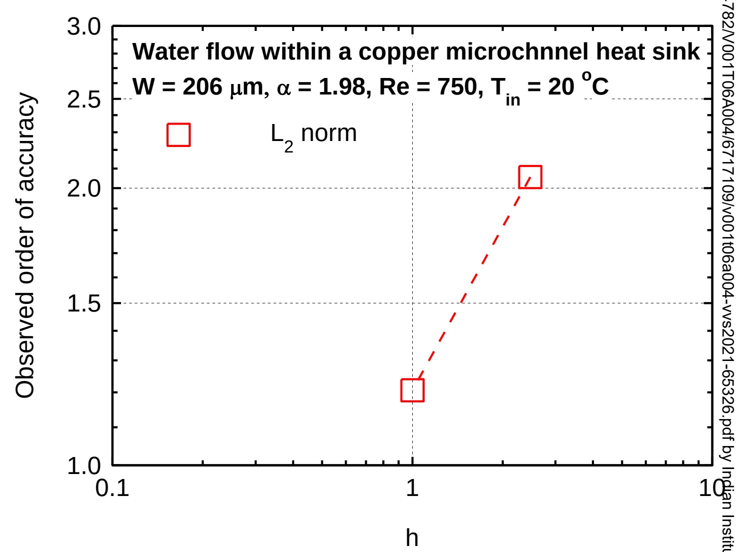

FIGURE 8: OBSERVED ORDER OF ACCURACY FOR WATER FLOW WITHIN A COPPER MICROCHANNEL WITH ASPECT RATIO OF 1.98 AND REYNOLDS NUMBER OF 800

Numerical uncertainty and grid convergence index (GCI) for obtained local Nusselt numbers using the proposed numerical procedure is estimated using the equation 7. The SRQ in this analysis is the temperature which is used to calculate the local and average value for Nu.The applied global deviation GCI process in this work can be found in [16–20].

$$GCI = \frac{F_s}{r^p - 1} |\varepsilon_h| \;\; ; U_{DE} = F_s |\varepsilon_h| \tag{7}$$

Where, F_s is the safety factor, p is the observed order of accuracy and ε_h is the discretization error. Safety factor, F_s, is considered as 3 for this work since the observed order of accuracy in Figure 8 is lower than the formal order of accuracy. GCI is estimated as 0.438 for obtained Nusselt numbers within the investigated microchannel heat sink while the numerical uncertainty is estimated to be as 0.13.

4. CONCLUSION

Numerical simulations are performed to study water flow and heat transfer within small-scale heat sinks. Obtained numerical results was compared with experimental data to complete validation process. Good agreement between numerical results, experimental data verified that the proposed numerical procedure was reliable to investigate heat transfer within miniature heat sinks. Several different mesh grid configurations were generated to obtain observed order of accuracy and complete solution verification process. The implemented numerical procedure had the observed order of accuracy of 1.2 in case of conjugate heat transfer within micro-scale heat sink. The observed order of accuracy for water flow within a microtube with uniform heat flux on the other hand, was observed to be 1.81. The estimated numerical uncertainty for obtained numerical results of Nusselt number was estimated as 0.13.

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